Subject: Mathematics | **PERIODIC TEST – 06 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **Max. Marks:** 30

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

Question 1 [1 Mark]

A girl calculates that the probability of her winning the first prize in a lottery is 0.08. If 6000 tickets are sold, how many tickets did she buy?

(a) 480 (b) 48
(c) 240 (d) 600

Working & Calculation:

- Let the number of tickets bought by the girl be x.
- Total tickets sold in the lottery $n(S) = 6000$.
- Probability formula:

$$P(\text{winning first prize}) = \frac{x}{n(S)} = \frac{x}{6000} = 0.08$$

- Solving for x:

$$x = 6000 \times 0.08 = 480 \text{ tickets}$$

CORRECT ANSWER: OPTION (a) ⟶ 480 tickets

Question 2 [1 Mark]

Savita and Hamida are friends. What is the probability that both will have different birthdays (ignoring a leap year)?

(a) $\frac{364}{365}$ (b) $\frac{1}{365}$
(c) $\frac{1}{7}$ (d) $\frac{365}{366}$

Working & Calculation:

- In a non-leap year, there are 365 possible days for a birthday.
- Savita's birthday can be on any of the 365 days of the year.
- For Hamida to have a different birthday, her birthday can be on any day except Savita's birthday:

$$\text{Favourable days for Hamida} = 365 - 1 = 364 \text{ days}$$

- Probability:

$$P(\text{different birthdays}) = \frac{364}{365}$$

CORRECT ANSWER: OPTION (a) ⟶ $\frac{364}{365}$

Question 3 [1 Mark]

A number is selected from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. The probability that it is a root of the quadratic equation $x^2 - 6x + 8 = 0$ is:

(a) $\frac{1}{4}$ (b) $\frac{1}{8}$

(c) $\frac{3}{8}$ (d) $\frac{1}{2}$

Working & Calculation:

- Sample space: $S = \{1, 2, 3, 4, 5, 6, 7, 8\} \implies n(S) = 8$.
- Solving the quadratic equation $x^2 - 6x + 8 = 0$:

$$x^2 - 4x - 2x + 8 = 0 \implies (x - 4)(x - 2) = 0 \implies x = 2 \text{ or } x = 4$$

- Both roots 2 and 4 belong to set $S \implies E = \{2, 4\} \implies n(E) = 2$.

$$P(\text{root of equation}) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{4}$

Question 4 [1 Mark]

In a spinning arrow game of chance, the wheel is divided into 8 equal sectors numbered 1 to 8. The probability that the pointer stops at an odd number greater than 3 is:

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$

(c) $\frac{3}{8}$ (d) $\frac{1}{8}$

Working & Calculation:

- Sample space: $S = \{1, 2, 3, 4, 5, 6, 7, 8\} \implies n(S) = 8$.
- Odd numbers strictly greater than 3 are:

$$E = \{5, 7\} \implies n(E) = 2$$

- Probability:

$$P(\text{odd number } > 3) = \frac{n(E)}{n(S)} = \frac{2}{8} = \frac{1}{4}$$

CORRECT ANSWER: OPTION (a) $\longrightarrow \frac{1}{4}$

Question 5 (Assertion–Reason) [1 Mark]

Assertion (A): In a bag containing 6 red balls and some blue balls, if the probability of drawing a blue ball is twice that of a red ball, then the number of blue balls is 12.

Reason (R): Let number of blue balls be x. Then $\frac{x}{6+x} = 2 \cdot \frac{6}{6+x} \implies x = 12$.

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

Step-by-Step Logical Analysis:

1. **Reason (R):** Red balls = 6, blue balls = $x \implies$ Total = $6 + x$. Given $P(\text{blue}) = 2 \times P(\text{red}) \implies \frac{x}{6+x} = 2\left(\frac{6}{6+x}\right) \implies x = 12. \implies$ **(R) is TRUE.**
2. **Assertion (A):** Number of blue balls is indeed 12. $\implies$ **(A) is TRUE.**
3. **Explanation Check:** The mathematical formulation and solution in (R) directly derive and explain (A).

CORRECT ANSWER: OPTION (a) $\longrightarrow$ Both (A) and (R) are true, and (R) is the correct explanation of (A).

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

Question 6 [2 Marks]

A bag contains lemon flavoured candies only. Malini takes out one candy without looking into the bag. What is the probability that she takes out: (i) an orange flavoured candy? (ii) a lemon flavoured candy?

Analysis of Candies:

(i) An orange flavoured candy: The bag contains lemon candies only; there are zero orange candies. This is an impossible event:

$$P(\text{orange flavoured candy}) = 0$$

Answer (i): P(orange flavoured candy) = 0

(ii) A lemon flavoured candy: Every candy in the bag is lemon flavoured. This is a sure (certain) event:

$$P(\text{lemon flavoured candy}) = 1$$

Answer (ii): P(lemon flavoured candy) = 1

Question 7 [2 Marks]

Two dice are thrown at the same time. Find the probability of getting: (i) a doublet (ii) a sum of 10.

Sample Space: Total elementary outcomes: $n(S) = 6 \times 6 = 36$.

(i) A doublet: Doublets are $\{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\} \implies n(E_1) = 6 \implies P = \dfrac{6}{36} = \dfrac{1}{6}$.

Answer (i): P(a doublet) = $\dfrac{1}{6}$

(ii) A sum of 10: Pairs summing to 10 are $\{(4,6),(5,5),(6,4)\} \implies n(E_2) = 3 \implies P = \dfrac{3}{36} = \dfrac{1}{12}$.

Answer (ii): P(sum of 10) = $\dfrac{1}{12}$

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

Question 8 [3 Marks]

A box contains 100 red cards, 200 yellow cards, and 50 blue cards. If a card is drawn at random from the box, find the probability that it will be: (i) a blue card (ii) not a yellow card (iii) neither yellow nor blue.

Given Data: Total cards $n(S) = 100\ (\text{red}) + 200\ (\text{yellow}) + 50\ (\text{blue}) = 350$ cards.

(i) A blue card: Number of blue cards $= 50 \implies P = \dfrac{50}{350} = \dfrac{1}{7}$.

Answer (i): P(blue card) = $\dfrac{1}{7}$

(ii) Not a yellow card: Cards other than yellow $= 350 - 200 = 150 \implies P = \dfrac{150}{350} = \dfrac{3}{7}$.

Answer (ii): P(not a yellow card) = $\dfrac{3}{7}$

(iii) Neither yellow nor blue: Must be red $\implies$ 100 cards $\implies P = \dfrac{100}{350} = \dfrac{2}{7}$.

Answer (iii): P(neither yellow nor blue) = $\dfrac{2}{7}$

Question 9 [3 Marks]

In a bag there are 44 identical cards with figures of circle or square on them. There are 24 circles, of which 9 are blue and 15 are green. There are 20 squares, of which 11 are blue and 9 are green. One card is drawn at random. Find the probability that it has: (i) a square (ii) a green circle (iii) a blue figure.

Card Inventory: Total $= 44$. Circles $= 24$ (9 blue, 15 green). Squares $= 20$ (11 blue, 9 green).

(i) A square: Number of squares $= 20 \implies P = \dfrac{20}{44} = \dfrac{5}{11}$.

Answer (i): $\mathbf{P(\text{a square}) = \dfrac{5}{11}}$

(ii) A green circle: Green circles $= 15 \implies P = \dfrac{15}{44}$.

Answer (ii): $\mathbf{P(\text{a green circle}) = \dfrac{15}{44}}$

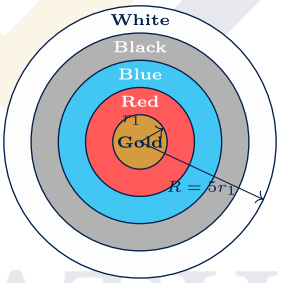
(iii) A blue figure: Blue circles (9) + Blue squares (11) $= 20 \implies P = \dfrac{20}{44} = \dfrac{5}{11}$.

Answer (iii): $\mathbf{P(\text{a blue figure}) = \dfrac{5}{11}}$

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [10 Marks]

Question 10 (Archery Concentric Scoring Rings) [5 Marks]

An archery target consists of five concentric circular scoring rings marked Gold, Red, Blue, Black, and White from the center outwards. The diameter of the central Gold region is 21 cm and each of the other bands is 10.5 cm wide. An arrow hits the target at random. Taking $\pi = \dfrac{22}{7}$:



Radii Setup: $r_1 = 10.5 \text{ cm} = \dfrac{21}{2}$ cm. Area unit: $A_1 = \pi r_1^2 = \dfrac{22}{7} \times \left(\dfrac{21}{2}\right)^2 = 346.5 \text{ cm}^2$.

(i) Area of Gold region: $A_{\text{Gold}} = A_1 = 346.5 \text{ cm}^2$.

Answer (i): $\mathbf{A_{Gold} = 346.5 \text{ cm}^2}$

(ii) Total area of entire target ($R = 52.5$ cm $= 5r_1$): $A_{\text{total}} = 25 \times A_1 = 25 \times 346.5 = 8662.5 \text{ cm}^2$.

Answer (ii): $\mathbf{A_{total} = 8662.5 \text{ cm}^2}$

(iii) Probability of hitting Gold region: $P(\text{Gold}) = \dfrac{A_1}{25A_1} = \dfrac{1}{25} = 0.04$.

Answer (iii): $\mathbf{P(\text{Gold}) = \dfrac{1}{25} = 0.04}$

(iv) Area and probability of Red region: $A_{\text{Red}} = (2^2 - 1^2)A_1 = 3 \times 346.5 = 1039.5 \text{ cm}^2 \implies P = \dfrac{3}{25} = 0.12$.

Answer (iv): $\mathbf{A_{Red} = 1039.5 \text{ cm}^2}$, $\mathbf{P(\text{Red}) = \dfrac{3}{25} = 0.12}$

(v) Ratio of P(Gold) to P(White): $A_{\text{White}} = (5^2 - 4^2)A_1 = 9A_1 \implies \text{Ratio} = \dfrac{1/25}{9/25} = \dfrac{1}{9}$.

Answer (v): $\mathbf{\text{Ratio } P(\text{Gold}) : P(\text{White}) = 1 : 9}$

Question 11 (Marble Mixture Analysis) [5 Marks]

A jar contains red, white, and blue marbles. The probability of selecting a red marble is $\frac{1}{4}$ and that of selecting a white marble is $\frac{1}{3}$. The jar contains 10 blue marbles.
(i) Find the probability of selecting a blue marble.
(ii) Find the total number of marbles in the jar.
(iii) Find the number of red marbles and white marbles in the jar.
(iv) If 4 red marbles are removed and 2 blue marbles are added, what is the new probability of drawing a red marble?

Given Probabilities: $P(\text{red}) = \frac{1}{4}$, $P(\text{white}) = \frac{1}{3}$, Blue marbles count $= 10$.

(i) Probability of selecting a blue marble [1 Mark]

Since the jar contains only red, white, and blue marbles, the sum of their probabilities equals 1:

$$P(\text{blue}) = 1 - [P(\text{red}) + P(\text{white})] = 1 - \left[\frac{1}{4} + \frac{1}{3}\right] = 1 - \frac{7}{12} = \frac{5}{12}$$

Answer (i): $\mathbf{P(\text{blue marble}) = \frac{5}{12}}$

(ii) Total number of marbles in the jar [1.5 Marks]

Let the total number of marbles in the jar be N:

$$P(\text{blue}) = \frac{\text{Number of Blue Marbles}}{N} \implies \frac{5}{12} = \frac{10}{N}$$

Cross-multiplying gives:

$$5N = 10 \times 12 = 120 \implies N = \frac{120}{5} = 24 \text{ marbles}$$

Answer (ii): N = 24 total marbles

(iii) Number of red marbles and white marbles [1.5 Marks]

Using the total count $N = 24$:

$$\text{Red Marbles} = P(\text{red}) \times N = \frac{1}{4} \times 24 = 6 \text{ marbles}$$

$$\text{White Marbles} = P(\text{white}) \times N = \frac{1}{3} \times 24 = 8 \text{ marbles}$$

Verification: 6 (red) + 8 (white) + 10 (blue) = 24 marbles.

Answer (iii): Red Marbles = 6, White Marbles = 8

(iv) New probability after removing 4 red and adding 2 blue marbles [1 Mark]

After the modifications:

- New number of red marbles $= 6 - 4 = 2$
- New number of blue marbles $= 10 + 2 = 12$
- Number of white marbles $= 8$
- New total marbles $N' = 2 + 12 + 8 = 22$ marbles (or $24 - 4 + 2 = 22$)

$$P'(\text{red}) = \frac{\text{New Red Marbles}}{N'} = \frac{2}{22} = \frac{1}{11}$$

Answer (iv): $\mathbf{P'(\text{red}) = \frac{1}{11}}$

— END OF WORKED SOLUTIONS (TEST–12) · MATHS MANTHAN TUITION CLASSES —